

LIRUI WANG

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EDUCATION

Massachusetts Institute of Technology

Sep 2021 -

University of Washington, Seattle

Sep 2016 - Mar 2021

Bachelor of Science, Electrical Engineering (Honors), College of Engineering

Bachelor of Science, Computer Science (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics (Minor), College of Arts and Sciences

Master of Science, Computer Science, College of Arts and Sciences

summa cum laude

Advisor: Professor Dieter Fox

RESEARCH PROJECTS

6D Instance Grasping with Point Cloud

- Investigated actor-critics methods with planner demonstrations and goal auxiliary tasks, to learn closed-loop control policies for 6D robotic grasping using point clouds.
- Achieved over 90% success rates on grasping various unseen objects in both simulation and real world.

Joint Motion and Grasp Planning

- Proposed an optimization-based integrated planner that optimizes manipulation trajectory with on-line grasp synthesis and selection.
- Demonstrated robust and smooth motion plans for grasping in cluttered scenes in the real world and outperformed several fixed-endpoint baselines.

Pose Tracking with the Deep Iterative Matching Network

- Evaluated DeepIM on YCB Video dataset and improved training on synthetic data by generating diverse pose distribution and realistic occlusions.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and AR applications.

WORK EXPERIENCE

Seattle, WA

Applied Research Intern, NVIDIA

- Investigated model-based RL for autonomous driving with a multi-modal RNN traffic predictive model from the bird's eye view.
- Used the predictive model latent to simulate reactive traffic and to learn adaptive cruise control with cut-in avoidance and adopted the learned model for long-horizon motion planning.

Seattle, WA

Teaching Assistant, UW CSE Department

- Assisted teaching for computer vision (CSE 455), deep learning (CSE 543), and artificial intelligence (CSE 473) at UW CSE department.
- Held office hours, helped and graded homework, and prepared course materials.

Shenzhen, China

Research Engineer Intern, DJI Technology

- Implemented functions such as take-off, tag detection, and multi-robot communication for drones.
- Applied SLAM to mapping and localization, and built a state machine to generate high-level plans.

MISCELLANEOUS PROJECTS

- **ARUW Robomaster Competition:** Managed a team of 10 students to design computer vision algorithms for an autonomous robot battle, including robots with onboard NVIDIA TX1 and drones with DJI Manifold with Guidance, in the RoboMasters competition hosted by DJI.
- **Unsupervised Flow and Depth Methods Study:** Studied structure from motion methods that jointly learn 3D vision tasks such as depth and flow.
- **Baxter Hand-Eye Coordination:** Developed software including camera models, hand-eye calibrations, robot kinematics and dynamics, shader-based rendering for Baxter to manipulate objects.
- **Microsoft Server Mover Project:** Built an autonomous mobile manipulator, a UR5 arm mounted on a server forklift, to pull out servers from a server rack.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification:** Investigated example meta-reweighting and adversarial training techniques to robustify a fossil classification task and surveyed on attacks and defense of adversarial examples for neural networks.
- **Survey on Continuous Trajectory Optimization:** Studied formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference on a factor graph using Gaussian Process.
- **Survey on Representation Learning for RL:** Study works that study conditions and benefits of representation for improving RL sample complexity and robotic applications of representation RL.

HONORS

- MIT EECS Fellowship (Xianhong Wu Fellow), MIT 2021 - 2022
- Herbold Data Science Fellowship, UW 2020 - 2021
- Patricia Lynch & Eugene Russell Memorial Computer Science Scholarship, UW 2018 - 2019
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship, UW 2017 - 2018
- Tau Beta Pi Scholarship, UW 2017 - 2019
- Annual Dean's List, UW, 2016 - 2020

PUBLICATIONS

- **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
Lirui Wang, Yu Xiang, Dieter Fox
Under Review
- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
Lirui Wang, Yu Xiang, Dieter Fox
In Proceedings of Robotics: Science and Systems (RSS), 2020

SELECTED COURSES

- **UW:** Deep Learning, Computer Vision, Computer Graphics, Modern Algorithm Toolbox, Robustness in Machine Learning, Interactive Learning, Probabilistic Robotics, Reinforcement Learning, Data Science, Theory of Deep Learning, Control Theory, Optimization and Learning for Control, Convex Optimization and Modeling, Numerical Analysis, Convex Algorithm and Analysis, High Dimensional Statistics, Statistical Learning